

Interactive Digital Art and Design SE608

Project Proposal Year IV



Student Name:	Montana Nicholas	Student Number:	C00297005
Working Title:	RATS!		
Description:	RATS! is an original strategic tabletop game set inside a house that has become a battlefield between a colony of rats and a household cat or cats. Players take on the role of either the rats, who must navigate through the house and ultimately reach the fridge in the kitchen, or the cats, who must prevent them from doing so. The game will feature a tile based board representing different areas of the house, including the kitchen, living room, bedroom, bathroom and connecting hallways. Rats will be able to use hidden routes such as cracks, vents and pipes to navigate the house, while cats can move throughout the board, place traps and pursue the rats. Players will use dice for movement, while cards will provide different abilities, attacks, healing and other gameplay actions. Combat between cats and rats will incorporate card based mechanics and dice based probability. These gameplay systems will be developed, prototyped and refined through research and playtesting throughout the project. The physical game will be the primary outcome of the project. This will involve designing the game's visual identity, characters, environments, board, cards, tokens, traps and other components through 2D concept art, 3D modelling and 3D printing. The project will also include the design and production of packaging and documentation for the completed game. Alongside the physical game, an interactive digital companion will be developed to work alongside the physical game. Rather than functioning as a separate digital version of the game or simply as a website containing the rules, the companion will provide interactive tools that can be used during a physical game. The digital companion will include features such as an interactive game setup guide, digital dice roller, card drawing/shuffling system and player health tracking. Physical game components may also be connected to the digital companion through methods such as QR codes or, where technically feasible, NFC tags. These could allow players to access information about individual characters, cards or game components by interacting with them through a phone or other device. Three.js may also be explored to allow players to view and interact with 3D		

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	representations of the game's characters, components or house environment. The digital component will be developed alongside the physical game, with its features being decided and refined as the physical game is developed and tested
Reasons for selecting project:	I am interested in developing this project because it combines several areas of Interactive Digital Art and Design that I enjoy, particularly character design, 3D modelling, visual design and physical prototyping. I am interested in creating a physical product rather than only a digital experience, and I would like the final outcome to feel like a complete, professionally designed game rather than a collection of individual assets. This project will allow me to develop my skills in areas that I have previously only worked with briefly at an introductory level, particularly in 3D modelling, 3D printing and web development. I am also interested in the process of designing a game from the initial concept through to a playable physical prototype. This will allow me to experiment with gameplay mechanics, test the game with players and make changes based on feedback throughout the development process. The project will also allow me to explore how physical and digital media can work together within one interactive experience. The physical board game will remain the main focus, while the digital companion will provide additional interactive functionality. This will allow me to incorporate programming, web development and UI/UX design into the project without making the digital application a replacement for the physical game
Proposed research content:	Tabletop Game Design My research will focus on the design and development of tabletop games, particularly strategic and asymmetric games where different players have different objectives, abilities and methods of playing. My research will also investigate how movement, randomness, resource management, health systems, combat, hidden information and player interaction can be combined to create an engaging and balanced game. Game mechanics will be researched and tested throughout development, with playtesting being used to identify problems with the game's balance, rules and overall player experience.

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Board Game Mechanics

Reference games will be researched mainly for their gameplay systems, including movement, combat, territory control, asymmetric abilities, resource management, cards, dice and player interaction.

- **Risk:** territory control, movement and strategic decision making
- **Cluedo:** hidden information and exploration
- **Root:** asymmetric factions, different abilities and objectives, and territory control
- **King of Tokyo:** dice based gameplay, health and combat

These games will be used as references for individual mechanics rather than as direct templates for the overall game.

Visual Design

Visual references will be researched for their visual identity, character design, illustration style, colour and presentation. These references are mainly relevant to the playful, cartoon-like visual style I would like to develop for **RATS!**

- **My Little Scythe:** colourful environments, simplified character designs and a playful interpretation of strategic gameplay
- **Exploding Kittens:** simple character illustrations, humour and card design
- **Sushi Go!:** simple illustrations, colour coding and clear card layouts
- **Mice & Mystics:** storybook style character design and representation of a large environment from the perspective of small characters

Physical Design and 3D Printing

My research will investigate the design and production of physical tabletop game components using 3D modelling and 3D printing. This will include research into scale, ergonomics, durability, printability, assembly and material considerations.

3D modelling will also be investigated as part of the overall visual development process, allowing characters, environmental objects and game components to be designed specifically for both visual presentation and physical production.

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Digital Companion and Web Development

My research will investigate how digital technology can be used to support a physical tabletop game without replacing the physical experience. Existing physical/digital hybrid games will be examined to explore how digital systems can interact with physical game components and assist with areas such as game setup, rules, information, randomisation and player tracking.

My research will also investigate interactive web design and UI/UX, including navigation, information architecture, responsive design and presenting game information clearly.

Svelte, HTML, CSS and JavaScript will be used as the primary technologies for developing the digital companion. Three.js may be explored for displaying and interacting with 3D game assets within the browser.

Different methods of connecting physical components to the digital companion will also be investigated. This may include QR codes, visual markers or NFC technology, depending on technical feasibility. NFC will be treated as an area for investigation and experimentation rather than a required feature of the final product

Physical and Digital Interaction

The physical board game and digital companion will be designed as two connected parts of the same game experience.

The physical board, characters, cards and other components will remain central to gameplay, while the digital companion will provide additional interactive functions rather than replacing the physical game.

For example, players may use the digital companion to:

- Set up a new game through an interactive setup guide
- Roll digital dice when required
- Draw and shuffle digital cards
- Track the health of rats and cats
- View information about characters, cards and traps
- Scan or interact with specific physical game components to display additional information
- View 3D models of characters or game components
- Access the rules and instructions during gameplay

QR codes or similar identifiers could be incorporated into selected physical components so that players can interact with them using a phone or tablet. NFC may also be investigated as a possible method of interaction where technically feasible.

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The aim is for the digital component to add functionality to the physical game, rather than simply acting as an online rulebook.

Technical Development

The main technical direction of the project will be web based development using Svelte, JavaScript, HTML and CSS. This approach allows the digital companion to be accessed through a browser on a computer, tablet or mobile device without requiring the player to install a separate application.

The digital companion will use JavaScript to implement interactive functionality such as dice rolling, card randomisation, health tracking and game-state information.

Three.js will be investigated as a possible way of displaying the game's 3D models within the browser. 3D models created in Blender or Maya could potentially be exported into a web-compatible format and displayed as interactive assets within the digital companion.

Unity and C# will be considered as alternative technical approaches, particularly for more complex interactive 3D functionality. However, the current proposed approach is to use web technologies because they are more appropriate to the intended role and scale of the digital companion.

Godot may also be used for small scale technical experimentation or prototyping if required, particularly where it can support exploration of interactive systems. It is not currently intended to become the main development environment for the project

Role of the Digital Component

The digital companion will form a secondary but technically significant outcome of the project.

The physical board game will remain the primary outcome, including the board, characters, cards, traps, tokens, packaging and other physical components. The digital companion will demonstrate the integration of programming, web development, interaction design and 3D content alongside the physical product.

The final digital component will aim to demonstrate functional interactive features rather than simply presenting information. The exact features will be refined throughout development based on the needs of the physical game.

The intended final functionality includes:

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	1. Interactive game setup 2. Digital dice roller 3. Card drawing/shuffling system 4. Player health tracking 5. Character and card information 6. Interactive rules and instructions 7. Physical-to-digital interaction through QR codes or potentially NFC 8. Interactive 3D game assets using Three.js, if technically achievable Not all proposed features will necessarily be included in the final version. Development will prioritise the features that provide the most useful interaction with the physical game while remaining achievable within the project timeframe. Playtesting and Research Ethics Playtesting will be used throughout development to evaluate the usability, balance and enjoyment of the physical game and digital companion. Playtesting will initially be carried out with classmates and peers during development. If external participants outside the normal student/class setting are required, the relevant SETU research ethics requirements will be reviewed and followed before testing begins. Playtesting sessions will be used to identify issues with movement, combat, card mechanics, game balance, physical components and the usability of the digital companion. Feedback will then be used to refine the project.
External links (if applicable):	Game Design / Board Game Research • A data driven review of board game design and interactions of their mechanics, Samarasinghe et al. (2021) Link • Don't Flip the Table Yet: A Framework for the Analysis of Visual and Cognitive Ergonomics in Board Games - Passarelli et al. (2024) Link • Meeple Centred Design: A Heuristic Toolkit for Evaluating the Accessibility of Tabletop Games - Heron et al. (2018) Link Physical / 3D Printing • Current and Future Opportunities for 3D Printing in Modern Board Games - Simões (2023) Link

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	Digital/ Hybrid Games - Hybrid Board Game Design Guidelines - Kankainen & Paavilainen (2019) Link - Examining the Digital in Hybrid Board Games - University of Melbourne Link Technology - Three.js Documentation Link
Hardware requirements:	- Personal computer/laptop to run 3D modelling and development software 3D printer - Filament/resin and other 3D printing materials - Materials for physical board and prototype construction - Cutting/Assembly tools as required - Tablet, phone, laptop etc to test the digital companion - Optional NFC tags/readers if NFC experimentation is undertaken
Software requirements:	Maya/Blender: for 3D modelling Svelte, HTML/CSS/JavaScript: web development Visual Studio Code: programming and development GitHub Penpot: UI/UX planning and prototyping Krita: concept art and any visual assets Three.js: potentially used for interactive 3D components on the website 3D printer slicing software?: prepping models for 3D printing Godot: optional technical prototyping
Other requirements:	Play testers: I would need participants for testing and evaluating the physical board game throughout the development
Changes Made / Actions Taken	Following feedback on the initial proposal, the role of the digital component has been expanded so that it forms a more active part of the physical board game experience rather than functioning mainly as a supporting website. The revised proposal now defines several possible interactions between the physical and digital components, including digital dice rolling, card drawing and shuffling, player health tracking, interactive setup and the use of QR codes or NFC to connect selected physical components to digital content. The technical direction has also been clarified. Svelte, JavaScript, HTML and CSS will be used as the primary

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	technologies, with Three.js being investigated for interactive 3D content. Unity/C# has been considered as an alternative technical approach following the feedback received, but the current planned direction is to use Svelte, JavaScript, HTML and CSS. Godot may also be used for smaller scale technical experimentation, if appropriate. The digital companion has been identified as a secondary but technically significant outcome, while the physical board game remains the primary outcome. Playtesting and the relevant research ethics requirements have also been considered and added to the proposal.
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Signed:
MONTANA NICHOLAS

Date:
24/09/2026

For Office Use Only

Approved/Not Approved:		
Reasons for not approving project:		
Conditions attached to approving project:		
Approved/Not Approved:		
Name of Supervisor:		
Signed: _____		Date: _____